

Briefcase: AI Transparency Proposal

Fall 2026 Senior Project

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I. Description

A. High-Level Overview

The Innocence Project works to exonerate people who have been wrongfully convicted of a crime that they did not commit. The Loyola Law School is a part of this work in the Los Angeles area through a project called Project for the Innocent (LPI). The Briefcase project is an “intelligent legal case management system” that works in collaboration with LPI. Court trial transcripts are thousands of pages long and are hard for attorneys to process. Briefcase’s purpose is to help the attorneys at LPI process court cases and the assorted documents that come with them faster, more efficiently, and more easily than they could by hand, expediting the number of cases they’re able to take on a year and the amount of people they can help.

B. Software Requirements

1. Frontend: React Typescript
2. Backend: Python, Flask, Firebase, ElasticSearch, AWS RDS, Postgres
3. NLP/ML:
 - a) Text Extraction: Tesseract OCR + ScikitLearn
 - (1) Clustering system to accurately extract text
 - b) Structure + Relationship: SpaCy
 - (1) POS Tagging, NER, Rule-Based Matching
 - c) Higher-level question answering: PyTorch, Sklearn, Huggingface

C. Target Use Case/Problem

Up until this point, Briefcase’s features have existed for the lawyers to use and the developers to understand, but we believe there can be more transparency from a developer-to-user standpoint— where their answers are being sourced from, clear citations, and a measure of confidence that attorneys can take into account when using Briefcase. When a subtle hallucination or text recognition mistake can make the difference between someone being exonerated or not, accuracy and complete confidence in the answers you’re receiving are of the utmost importance.

D. Features

Currently, the entire system is dependent on the text extraction, which makes subsequent dependencies less reliable and accurate. To improve transparency, we plan to implement better information referencing and measure the confidence of the following features:

1. OCR
 - a. We plan to flag areas from the source document where the text recognition is less confident in its estimate using confidence scoring so that its users can more carefully examine the transcript.
2. Minimap Summary
 - a. We plan to add a confidence score or indicator for each AI-generated summary within the minimap feature built-in to each transcript, as well as a comprehensive citation system for sourcing its facts to cross-reference it with the original source, improve its accuracy, and reduce the likelihood of hallucinations.
3. DocChat
 - a. Briefcase allows attorneys to ask a DocChat questions about the case, but it is prone to producing answers without demonstrating how reliable they are, or where they are sourced from. We plan to create a more structured way of attaching citations to generated answers so attorneys can see where the information came from in the original trial transcript.

II. Justification

While Briefcase deals extensively with ML and NLP, areas that our group has had little exposure directly working with, this project provides a great opportunity to not only continue on the legacy of such impactful work but also become familiar with the ML and NLP fields, which are especially relevant in today's software industry.